
A-Worthy Education

H2 Economics 9570 Study Suite

Document 57

Macro CSQ Set D — Model Answers

A-Grade Student-Script Style with Examiner Notes

Contents

Case Study D1 — Model Answers

(a) Two features of China's macroeconomic position [2 marks]

Worked Example

(i) Growth at 5.0% is well below the historical 9.5% average, indicating a structural slowdown; (ii) the divergence between manufacturing PMI at 49.8 (contraction) and services PMI at 51.2 (expansion) indicates that the slowdown is concentrated in the industrial sector.

(b) Effect of property correction on AD [6 marks]

Worked Example

The property sector affects AD through four channels.

Investment. Property investment fell 9.6%, directly reducing the I component of AD. Construction-related investment also fell through linkages to materials and machinery sectors.

Consumption (wealth effect). Property accounts for 60–70% of urban household wealth. Falling prices reduce perceived wealth, lowering consumption — particularly of big-ticket items (autos, appliances, services). Retail sales growth at 3.5% versus pre-pandemic 8–10% reflects this.

Local government revenue. Land sales account for 30–40% of local government revenue. Falling land values compress this revenue, forcing cuts to local infrastructure spending — a fiscal-cum-investment contraction.

Confidence and credit. Falling property prices and developer defaults damage household confidence, raising savings rates and reducing consumption. They also impair bank balance sheets through non-performing loans, tightening credit availability and reducing investment further.

The cumulative effect shifts AD leftward materially, consistent with low headline CPI (0.2%) and negative PPI (−2.2%).

(c) Deflation risk in China [6 marks]

Worked Example

Concept. Deflation is a sustained fall in the general price level. It is harmful because: (i) it raises the real burden of debt, (ii) it creates incentives to postpone spending (deflationary spiral), (iii) it limits the effectiveness of monetary policy when nominal rates are near zero.

Evidence of deflation risk. CPI at 0.2% is barely positive. PPI at −2.2% YoY shows factory-gate deflation. The combination indicates demand weakness across both consumer and producer markets. Property prices are falling; corporate margins are squeezed; wage growth is slowing.

Mitigating factors. CPI remains positive, distinguishing China from Japan in the 1990s. Food prices have a large weight and are typically volatile; core inflation excluding food is closer to 0.5–1%. Policy stimulus (PBOC cuts, fiscal expansion) is being deployed.

Judgement. China faces a genuine deflation risk, comparable to Japan's late-1990s position. The risk is highest if (i) property prices continue falling, (ii) consumer confidence

does not recover, and (iii) policy stimulus is insufficient. The current policy mix is responsive but, by historical comparison with effective stabilisations, on the modest side.

(d) PBOC monetary policy effectiveness [8 marks]

Worked Example

Modest impact achieved. The 50 bps RRR cut and 30 bps reverse-repo cut lower bank funding costs, supporting credit supply. Combined with the property-purchase easing, this has slowed (though not arrested) the property correction. Equity markets have firmed.

Limitations of monetary transmission.

Liquidity trap dynamics. When confidence is low, households save additional liquidity rather than spending. The marginal propensity to consume falls, weakening the transmission from lower rates to AD. Similar dynamics constrained Japan's monetary policy in the 1990s.

Balance-sheet constraint. Property developers and local governments are debt-overhung. Lower rates help service existing debt but do not encourage new borrowing if balance sheets are impaired. The transmission from policy rate to bank lending is therefore blunted.

Exchange-rate concerns. Aggressive easing would weaken the renminbi, conflicting with the PBOC's currency stability objective and risking capital outflows. This constrains how far rates can be cut.

Real rate stickiness. With CPI at 0.2%, the real interest rate is high even after nominal cuts. This dampens borrowing.

Judgement. PBOC policy is well-targeted but constrained in magnitude. Effectiveness is partial. The likely additional remedy is more aggressive fiscal expansion (the RMB 10 trillion debt swap is a start), supplemented by structural reforms (consumption tax cuts, social safety net expansion) that raise the propensity to spend.

(e) Youth unemployment — structural or cyclical? [8 marks]

Worked Example

Cyclical interpretation. The 17.1% youth unemployment rate has risen substantially from approximately 11% pre-pandemic. The recent rise coincides with the property correction and weakened domestic demand, suggesting a cyclical component. If aggregate demand recovers, hiring of young workers should rise.

Structural interpretation.

Education-job mismatch. China now produces over 11 million university graduates annually, but the bulk of new manufacturing and services jobs are mid-skill. Graduates queue for white-collar positions that grow slowly while blue-collar jobs go unfilled.

Sector composition. Tech and education sectors that historically absorbed graduates were hit by 2021–2022 regulatory crackdowns (after-school tutoring, fintech) and have not recovered. Property-related professional services (legal, finance, marketing) have contracted in tandem with the property sector.

Demographics and expectations. The one-child generation has high reservation wages and strong preferences for white-collar work — youth in some regions choose extended job search over accepting available work, raising measured unemployment.

Judgement. Both components are present, with structural factors larger than cyclical. The structural share is reflected in the fact that the rate began rising before the prop-

erty correction (early 2022 onwards). Cyclical recovery alone is unlikely to reduce youth unemployment below 12–13%; the remaining 4–5 percentage points require structural responses — vocational training, regional mobility incentives, expansion of the services and high-value-manufacturing job base.

Case Study D2 — Model Answers

(a) Two features of US macroeconomic position [2 marks]

Worked Example

(i) The US economy combined moderate growth (2.8%), low unemployment (4.2%), and inflation near target (2.9% headline) — an unusually benign combination after aggressive tightening; (ii) the fiscal deficit at 6.4% of GDP is high relative to the cyclical position (near full employment), suggesting a structural rather than cyclical deficit.

(b) Effect of Fed rate increase on AD [6 marks]

Worked Example

A 525 bps rise in the federal funds rate transmits to aggregate demand through four channels.

Investment channel. Higher policy rates raise corporate borrowing costs. Capital expenditure projects with internal rates of return below the new cost of capital are deferred or cancelled. Business investment growth slowed from 4.4% to 2.1% over the tightening cycle.

Consumption channel. Higher mortgage rates (from 3.0% to 7.5%) reduce affordability and slow housing demand. Higher credit-card and auto-loan rates reduce big-ticket consumption. Mortgage refinancing — historically a cash-out source for consumption — collapsed.

Wealth channel. Tighter rates initially depressed equity and bond prices, reducing household wealth. However, this channel was offset in 2024 as the S&P 500 rose 23% on AI-related earnings optimism.

Exchange-rate channel. Higher US rates attracted capital inflows, strengthening the dollar by approximately 10% during 2022–2023. This reduced net exports.

Collectively these channels shift AD leftward, reducing both inflation and (in principle) real output.

(c) Why avoiding recession [8 marks]

Worked Example

The combination of 525 bps tightening and 2.8% GDP growth in 2024 is historically anomalous. Four factors explain the resilience.

1. Fiscal offset. The federal deficit averaged 6.4% of GDP, well above the cyclical norm. The CHIPS Act and Inflation Reduction Act injected hundreds of billions into manufacturing and clean-energy investment. Fiscal expansion offset much of the monetary contraction at the level of aggregate demand.

2. Household balance-sheet strength. Households entered the tightening cycle with elevated savings from pandemic transfers and stimulus. Combined household net worth was at a record high. The wealth effect on consumption was positive even as rates rose.

3. Locked-in mortgage rates. Approximately 90% of US mortgages are 30-year fixed-rate. The rate hike did not transmit to existing mortgage payments. New buyers were locked out, but existing homeowners were protected — a unique buffer against the housing

channel of monetary transmission.

4. AI-driven investment surge. Capital expenditure on data centres, semiconductors, and AI infrastructure rose sharply through 2024, providing an investment boom that offset rate-driven contraction in other categories.

Limiting condition. These four buffers are partly one-off. Pandemic excess savings are depleting; mortgage refinancing demand will eventually rise; fiscal deficits face mounting interest-cost pressure. The soft landing is therefore partly a buffer-aided outcome rather than evidence that 525 bps of tightening is consistent with continued expansion.

(d) Sustainability of 6.4% deficit at full employment [6 marks]

Worked Example

Unsustainability concerns. A 6.4% deficit at near-full employment violates the textbook counter-cyclical fiscal rule. Federal debt rose to 99% of GDP, with interest expense approaching 3.5% of GDP. With 10-year Treasury yields at 4.4% and a positive term premium, debt service is rising faster than nominal GDP, implying an unstable debt trajectory absent policy adjustment.

Risks. (i) Crowding out — sustained large deficits at full employment raise real interest rates, reducing private investment over time. (ii) Bond-market reaction — if investors demand a higher term premium to absorb growing supply, yields rise, accelerating debt-service costs. (iii) Reduced fiscal space for the next downturn — without a buffer, recession response options shrink.

Mitigating factors. US dollar reserve-currency status provides exorbitant privilege — global demand for Treasuries is structurally high. The fiscal expansion includes growth-enhancing components (semiconductor capacity, clean-energy infrastructure) that may raise potential output.

Judgement. The 6.4% deficit is not sustainable indefinitely at full employment. It is sustainable for several more years given reserve-currency demand, but a path-correction (revenue increase or spending restraint) will eventually be required. The political-economy difficulty of either suggests the adjustment will come gradually and only when forced by bond-market conditions.

(e) Soft landing — genuine or deferred? [8 marks]

Worked Example

Genuine soft-landing case. Inflation has returned to within 1 percentage point of the 2% target. Unemployment at 4.2% remains low. Real GDP growth at 2.8% is at or above potential. The Fed has begun cutting rates from a peak that did not produce recession. Historical comparison: the 1994–1995 tightening cycle ended similarly, with stable expansion through the late 1990s.

Deferred-recession case. (i) Yield curve inversion (10-year minus 2-year) was sustained for over two years — historically the most reliable recession indicator. (ii) Pandemic excess savings have largely been depleted; the consumer cushion is fading. (iii) Commercial real estate stress (office vacancies, downtown valuations) has not fully crystallised — bank losses may be ahead. (iv) Long-and-variable lags: monetary policy effects can take 18–24 months to be felt; some of the 2023 tightening may bite in 2025–2026. (v) Fiscal stimulus is moderating from the IRA/CHIPS peak.

Composite assessment. The genuine vs. deferred distinction may be too binary. The

most plausible reading is that the US has executed an unusually well-timed soft landing on the cyclical dimension (inflation/output), while accumulating structural risks (debt, commercial real estate, fiscal trajectory) that may produce slower future growth without a discrete recession.

Judgement. On a 1–2 year horizon, the soft landing is genuine — recession risk is materially lower than 12 months ago. On a 3–5 year horizon, the unresolved fiscal and balance-sheet imbalances suggest growth will be slower than the recent 2.8% pace, with non-trivial recession risk at some point in the cycle. The soft landing is real but partial.

Case Study D3 — Model Answers

(a) Inflation versus hyperinflation [2 marks]

Worked Example

Inflation is a sustained rise in the general price level of an economy over time. Hyperinflation is an extreme and accelerating form of inflation, typically defined as exceeding 50% per month. Argentina's December 2023 monthly inflation of 25.5% approached but did not technically cross the hyperinflation threshold; annualised CPI at 211% represents very high inflation.

(b) Causes of Argentine inflation [6 marks]

Worked Example

Cause 1 — Monetary financing of fiscal deficits. Argentina's central bank historically created money to finance fiscal deficits — the classic “inflation tax”. With a fiscal deficit of 4.5% of GDP (2023) and limited access to external bond markets, the central bank financed the gap by money creation. The quasi-fiscal losses of the central bank (interest paid on absorption notes) themselves required more money creation, generating a self-reinforcing spiral.

Cause 2 — Exchange-rate collapse and pass-through. The official peso traded at multiple exchange rates, with the parallel “blue dollar” often double the official rate. Capital flight depleted reserves. Each devaluation passed through immediately to import prices and triggered domestic-price markups across the economy (Argentina's pass-through coefficient is among the highest globally). The December 2023 54% devaluation produced the 25.5% monthly inflation spike.

Cause 3 — Inflation expectations and indexation. With persistently high inflation, contracts, wages, and rents were widely indexed to past inflation. This created mechanical price-setting persistence — current inflation reflects past inflation regardless of underlying demand-supply balance.

(c) Costs of high inflation [6 marks]

Worked Example

Erosion of real incomes. Wages adjust with a lag; real wages fell 15% YoY (Extract D3.3). Lower-income households, who hold more wealth in cash and have less access to inflation-protected assets, suffer disproportionately. Poverty rate rose to 52.9%.

Distortion of price signals. When inflation is high and volatile, relative prices become noisy. Firms cannot distinguish a 30% rise in their inputs as a real price shift or as the general price-level rise — economic calculation deteriorates.

Discouragement of long-term investment. Investors and lenders demand large inflation risk premia. Long-term contracts become impossible. Productive investment is replaced by short-term speculation in currency, real estate, or inflation hedges.

Currency substitution and dollarisation. Argentines historically save in US dollars rather than pesos. This drains domestic savings from the formal banking system, raising borrowing costs for legitimate investment.

Fiscal vicious cycle. Inflation erodes the real value of tax revenue collected with a lag (Olivera-Tanzi effect), widening the fiscal deficit, which is financed by money creation, which raises inflation.

(d) Effectiveness of stabilisation [8 marks]

Worked Example

Achievements. The shift from 4.5% deficit to 0.4% primary surplus in a single year is exceptional. Quasi-fiscal money creation was eliminated. Monthly inflation fell from 25.5% (Dec 2023) to 2.7% (Oct 2024) — a near-tenfold deceleration. Country risk fell from over 2,000 bps to 759 bps, indicating restored market access. The exchange-rate gap between official and parallel rates narrowed.

Mechanisms of success. The classical stabilisation recipe (eliminate monetary financing, devalue, fiscal adjustment, anchor expectations) is exactly what was implemented. Removal of subsidies addressed both the deficit and relative-price distortions. Currency devaluation closed the parallel-rate gap.

Risks remaining. Inflation expectations remain high (12-month CPI at 193%); deindexation has not been formally engineered, so price persistence remains. FX reserves at US\$28bn are still low for a country of Argentina's size — vulnerable to a renewed run. The fiscal adjustment depended partly on inflation-eroded transfer payments, which may rebound if inflation falls fast enough that real values are restored.

Judgement. The stabilisation has been remarkably effective by historical standards (compare with Bolivia 1985, Brazil 1994 Real Plan). The remaining question is whether the 2024 disinflation can transition into stable single-digit inflation in 2025–2026, which depends on continued fiscal discipline and gradual exchange-rate normalisation. Early evidence favours optimism but the historical base rate for sustained Argentine stabilisation is poor.

(e) Social cost vs. benefit [8 marks]

Worked Example

Social cost. GDP contracted 3.5%. Real wages fell 15%. Poverty rose to 52.9% — over half the population. Public-sector layoffs (30,000 federal workers) and subsidy removal raised the immediate cost-of-living burden on lower-income households. The short-run distributional cost is severe and concentrated.

Social benefit. Disinflation from 25.5%/month to 2.7%/month substantially reduces the inflation tax on real wages and pension values going forward. The end of monetary financing means future deficits will not be paid for by money-creation-induced inflation. Restored access to bond markets reduces external borrowing costs and frees fiscal space.

Counterfactual. The relevant comparison is not stabilisation versus the pre-stabilisation status quo — it is stabilisation versus the trajectory that the pre-stabilisation path implied. Continued 200%+ inflation, with rising probability of true hyperinflation and a currency collapse, would have produced cumulative welfare losses far exceeding the 2024 contraction.

Distribution of costs. The benefits accrue over the medium run and are diffused; the costs are concentrated in 2024 and disproportionately on the bottom 50% of the income distribution. The political-economy challenge is to maintain consensus through

the adjustment period when costs are visible and benefits are not yet realised.

Judgement. The social cost in 2024 is real but lower than the implied counterfactual cost of inaction. Net welfare evaluation depends on weight placed on present versus future welfare and on willingness to accept short-run concentrated losses for long-run diffused gains. On reasonable welfare assumptions, the stabilisation is welfare-improving in expectation, conditional on it being sustained. The risk is that political pressure forces premature reversal before the medium-run benefits are crystallised.

Case Study D4 — Model Answers

(a) Two features of Vietnam's export sector [2 marks]

Worked Example

(i) Exports at 96% of GDP indicate extreme export dependence; (ii) sectoral concentration — electronics (35%) and textiles/footwear (20%) together account for 55% of exports — exposing the economy to sector-specific demand shocks.

(b) FDI and productive capacity [6 marks]

Worked Example

FDI inflows raise the productive capacity (LRAS) of the recipient economy through four channels.

Capital accumulation. FDI directly adds to the physical capital stock — factories, equipment, technology. Samsung's Vietnam operations alone exceed US\$22bn cumulative investment. Higher capital per worker raises labour productivity.

Technology transfer. Multinational subsidiaries import production technology and management practices that exceed the domestic frontier. Spillovers to local suppliers raise productivity in the broader economy, although the strength of spillovers depends on local absorptive capacity.

Skill formation. FDI-employing workers acquire firm-specific skills through training and on-the-job learning. As workers eventually move between firms, these skills diffuse through the labour market.

Backward and forward linkages. FDI plants create demand for local intermediates (suppliers) and outputs that local downstream firms use. The cluster effect can be substantial — Vietnam's industrial parks around Hanoi and Ho Chi Minh City are textbook cluster examples.

The combined effect shifts LRAS rightward, raising potential output and supporting non-inflationary growth. Vietnam's record FDI of US\$25.4bn in 2024 represents approximately 6% of GDP — a substantial annual addition to productive capacity.

(c) Implications of 50% domestic value-added [6 marks]

Worked Example

The fact. Vietnam's domestic value-added (DVA) share at 50% means half of export value reflects imported intermediates merely assembled in Vietnam. Vietnam captures only the assembly and labour margin, not the design, branding, or high-tech component manufacturing.

Implications for growth. Real GDP growth attributable to exports is roughly half what headline export growth suggests. The export-to-GDP ratio of 96% overstates the economic-significance share — the value-added contribution is closer to 48% of GDP. Vietnam's economic transformation is real but more modest than headlines imply.

Implications for vulnerability. Imported intermediates mean Vietnam is exposed both to upstream supply disruptions (semiconductor shortages, shipping disruptions) and to downstream demand swings. The country is a middle-link in global supply chains rather

than an autonomous producer.

Implications for development strategy. Climbing the value chain requires investment in upstream activities — local component manufacturing, design capability, R&D — that are not naturally part of an assembly-focused FDI base. Without active industrial policy, Vietnam risks the “middle-income trap” where assembly continues but value capture stalls.

Comparative perspective. South Korea’s DVA share is approximately 70%, China’s approximately 80%. Vietnam’s 50% is closer to the early stages of these countries’ development trajectories. The trajectory upward is technically achievable but requires sustained policy effort.

(d) Constraints to continued growth [8 marks]

Worked Example

Power grid. Power outages affecting industrial parks in 2023 and 2024 indicate that infrastructure has not kept pace with demand. Industrial customers cite reliability as a top-three concern. Grid investment requires multi-year capex programmes; in the interim, manufacturing growth may be physically capped by available power.

Industrial land costs. 16% rise in industrial land prices near Ho Chi Minh City in a single year erodes the cost advantage that drove FDI in the first place. If land costs continue rising at this pace, Vietnam’s labour-cost advantage may be partially offset by land-cost convergence with regional peers.

Skilled labour shortage. 78% of foreign investors citing skills as a binding constraint is striking. Vietnam’s transition from low-skill assembly (footwear, basic electronics) to higher-skill manufacturing (semiconductors, automotive) requires a workforce that the current education system does not yet supply at scale.

Wage convergence. Vietnamese manufacturing wages of US\$320/month versus China’s US\$830/month is a 2.6:1 ratio that has narrowed from 4:1 a decade ago. As wages rise (necessary for political legitimacy and worker welfare), the cost differential will continue narrowing. Vietnam’s competitive position will increasingly depend on productivity rather than wages alone.

Aggregate assessment. Power, land, and skills are simultaneously binding. The constraints are mutually reinforcing — without skills, productivity gains cannot offset rising labour and land costs; without power, additional FDI cannot be absorbed; without infrastructure, agglomeration economies cannot be realised.

Judgement. The constraints are real and material, but they are also the typical mid-development constraints of any economy at Vietnam’s per-capita income (around US\$4,300). Resolution requires sustained public investment in grid, transport, and education — exactly the agenda Vietnam’s current Five-Year Plan articulates. Implementation, not strategy, is the binding question.

(e) Supply-side policy for Vietnam [8 marks]

Worked Example

Case for supply-side policy. Vietnam’s binding constraints (power, land, skills) are explicitly supply-side. Each constraint is addressable by appropriate policy: grid investment, industrial-park expansion and rezoning, technical and vocational education expansion. These shift LRAS rightward and raise the absorptive capacity for FDI. Without

active supply-side policy, growth would slow as constraints tighten — diminishing returns to the existing FDI-driven model.

Specific supply-side measures.

Education and skills. Expanding technical universities and vocational training, with curricula co-designed with major investors (Samsung University, Foxconn programmes already operating). PCP-style mid-career conversion programmes for low-skill workers transitioning to high-skill segments.

Infrastructure. North-South high-speed rail, port expansion, doubling installed power capacity over the next decade including renewables (Vietnam has high solar potential).

R&D and technology absorption. Tax incentives for R&D, requirements for technology transfer in major FDI contracts, support for local supplier upgrading.

Trade and FTA network. Continued FTA expansion (Vietnam already has 17 in force) supports the export model and provides market access for any quality-upgrading the domestic economy achieves.

Risks of pure supply-side approach. Supply-side policy alone does not address the value-added question — the 50% DVA share may persist even with better infrastructure. Active industrial policy targeting upstream segments (component manufacturing, design) is needed, but carries the standard industrial-policy risk of picking losers.

Judgement. Vietnam should adopt supply-side policies as the central tool for sustaining its growth model. The constraints are explicitly supply-side and the appropriate response is supply-side investment in infrastructure, skills, and technology absorption. This should be complemented by a targeted (not pervasive) industrial policy aimed at moving up the value chain in priority sectors. The two work in tandem: supply-side investment creates the conditions; industrial policy shapes the direction. The combination is the standard high-growth-economy playbook (Korea, Taiwan, Singapore historically) and is the most credible path for Vietnam.

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